RESEARCH STATEMENT

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My research lies at the interface of **algebraic geometry, combinatorics, and number theory**, with a further strand in topology. A recurring theme is **configurations of points on spaces**: how can we classify or count these configurations? How can we describe the geometry and topology of the **moduli space** of these configurations? Depending on the rules for what happens when points collide, the moduli spaces I study include Hilbert and Quot schemes, moduli spaces of coherent sheaves with zero-dimensional support, and configuration spaces of distinct points.

My central long-term program is **counting zero-dimensional sheaves on singular curves**, with a focus on the **high-rank** setting (Section 1). It studies generating functions that count zero-dimensional sheaves with a framing—a choice of generators whose number is the rank—on singular curves. I pioneered this program through my study of unframed sheaves on the nodal curve [44], where I computed the relevant generating function and observed unusual rigidity. In work with Jiang (Theorem 2, [47]), I developed a systematic approach to these counts through higher-rank framings, established general properties, and computed initial examples demonstrating its effectiveness.

Sheaf counting is a recurring theme in enumerative geometry, a major meeting point of mathematics and theoretical physics: Donaldson–Thomas theory, for example, connects sheaf counting with string theory [64]. Singular curves bring a different set of connections into this picture. In rank one, the Oblomkov–Rasmussen–Shende conjecture relates Hilbert schemes of points on planar singular curves to knot theory [69]. Higher rank reveals phenomena even beyond this, and seeking explanations for them reveals new structures. A recent example is a new geometric framework yielding **Rogers–Ramanujan sum-product identities** proved in joint work with Lau and Ono (Theorems 1 and 4, [51]); this work is featured in *Scientific American* [22]. These identities arise from my geometric study of Quot schemes on singular curves of the form $X^a = Y^b$ with Jiang and Oblomkov (Theorem 3, [48]). The proof of these identities also reveals structural connections to cylindric partitions and compositional Dyck paths that were not identified in the original geometric work. This project also shaped my experience of collaborating with AI in mathematical research (Section 1.2.1).

There are also topological and arithmetic strands to this program. The varieties $C_n(X)$ of $n \times n$ "matrix points" on X provide concrete models for unframed sheaf counting. With Asvin G, Jiang, and Wei, I describe their rational cohomology rings for every complex variety X and obtain a uniform Macdonald-type formula for their Betti numbers (Theorem 5, [31]). This uniformity is striking in view of the subtle point-counting phenomena above. On the arithmetic side, Sato–Tate-type distributions of these point counts across special families of varieties led to joint work with Ono and Saad (Theorem 6, [54]) and an REU project that I co-mentored at the University of Virginia in 2023 [11].

My other programs are also guided by the search for structures behind rigid patterns. In the **topology of configuration spaces** (Section 2), my works with Cheong and with Ramos give rational generating functions for the mixed Hodge numbers of configuration spaces on punctured Riemann surfaces (Theorem 7, [16, 55]). I obtain a uniform formula describing how these invariants change when a point is removed from the underlying variety X, under suitable hypotheses on X [43]. Rationality as both the number of points and the cohomological degree vary suggests structures beyond the fixed-degree perspective of classical representation stability in Church–Ellenberg–Farb [23, 24].

In **random matrix theory** (Section 3), I study distributions of algebraic objects arising from integral and p-adic matrices, guided by Cohen–Lenstra heuristics and their random-matrix analogues [25, 78]. The key is to identify the richer structure of each model that makes the distribution law simple, or "Cohen–Lenstra". For products $M_1 M_2 \cdots M_k$ of independent Haar-random p-adic $n \times n$ matrices, I prove that the chain of

quotient maps $\mathrm{cok}(M_1 \cdots M_k) \twoheadrightarrow \ldots \twoheadrightarrow \mathrm{cok}(M_1)$ obeys the Cohen–Lenstra distribution [45], and its $n \to \infty$ limit is "universal" (Theorem 8, with Nguyen and Van Peski [53]). For a monic polynomial $P(t)$ and a Haar-random p-adic matrix M with prescribed residue, I show with Cheong that the law of $\mathrm{cok}(P(M))$ as a $\mathbb{Z}_p[t]/(P(t))$-module is Cohen–Lenstra subject to the modulo p constraint (Theorem 9, [17]).

In **arithmetic geometry** (Section 4), I study when algebraic constraints force rigidity. With Kadets and Martin, I show that all low-degree subvarieties varying algebraically over the universal family of sufficiently high-degree hypersurfaces in $\mathbb{P}^n$ arise from a tautological intersection construction (Theorem 10, [50]). Motivated by my work on quaternionic unit equations (Theorem 12, [42]), I prove finiteness theorems for ordered products in division algebras with Ghioca, and in matrix algebras with Bell and Ghioca, under suitable independence hypotheses (Theorem 11, [33, 7]).

Section 1 (pp. 2–6) develops my main program; Sections 2–4 (pp. 7–9) give a shorter account of my other research directions.

1. Zero-dimensional sheaves on singular curves

Matrix points give a concrete model for coherent sheaves with zero-dimensional support, or briefly, zero-dimensional sheaves. For $X \subseteq \mathbb{A}^d$ cut out by $f_1, \ldots, f_s$, its variety of $n \times n$ matrix points is

$$C_n(X) = \{(A_1, \ldots, A_d) \in \mathrm{Mat}_n^d : A_i A_j = A_j A_i,\ f_\ell(A_1, \ldots, A_d) = 0 \text{ for all } i, j, \ell\}.$$

An $n \times n$ matrix point is a zero-dimensional sheaf $\mathcal{Q}$ on X with an ordered basis of $H^0(X, \mathcal{Q})$, of dimension n. Forgetting this basis gives the moduli stack $\mathrm{Coh}_n(X) = [C_n(X)/\mathrm{GL}_n]$. Over $\mathbb{F}_q$, its *Coh zeta function* counts sheaves with the natural automorphism weight:

$$\widehat{Z}_X(t) = \sum_{n \geq 0} |\mathrm{Coh}_n(X)(\mathbb{F}_q)|\, t^n = \sum_{n \geq 0} \frac{|C_n(X)(\mathbb{F}_q)|}{|\mathrm{GL}_n(\mathbb{F}_q)|}\, t^n, \qquad |\mathrm{Coh}_n(X)(\mathbb{F}_q)| = \sum_{[\mathcal{Q}]} \frac{1}{|\mathrm{Aut}(\mathcal{Q})|}. \tag{1}$$

The sum is over isomorphism classes of zero-dimensional sheaves with $\dim_{\mathbb{F}_q} H^0(X, \mathcal{Q}) = n$; this definition also applies to nonaffine varieties. Even for $X = \mathbb{A}^d$, these spaces pose difficult geometric questions, especially for $d \geq 3$ [40, 56]; for example, determining for which n the variety $C_n(\mathbb{A}^3)$ is reducible remains an open problem, going back to Gerstenhaber [32]. Nevertheless, my initial calculations for singular curves [44] revealed unexpected regularity, leading to the following conjecture.

Conjecture 1 ([44])**.** *Let C be a reduced planar curve over $\mathbb{F}_q$, with normalization $\widetilde{C}$, and set $\widehat{N}_C(t) = \widehat{Z}_C(t)/\widehat{Z}_{\widetilde{C}}(t)$.*

(a) *The power series $\widehat{N}_C(t)$ has infinite radius of convergence* [44].
(b) *The special value $\widehat{N}_C(1)$ has a periodic product expression of some modulus $M \in \mathbb{Z}_{\geq 1}$: there is an M-periodic function $r : \mathbb{Z} \to \mathbb{Z}$ such that $\widehat{N}_C(1) = \prod_{j=1}^{\infty}(1 - q^{-j})^{-r(j)}$.*

The cusp $C :\ X^2 = Y^3$ illustrates the prediction. With Jiang [47], I prove

$$\widehat{N}_C(t) = \sum_{m \geq 0} \frac{q^{-m^2} t^{2m}}{(q^{-1}; q^{-1})_m};$$

here and below we use the q-Pochhammer symbols

$$(z; q)_m = \prod_{j=0}^{m-1} (1 - zq^j), \quad (z_1, \ldots, z_r; q)_m = \prod_{i=1}^{r} (z_i; q)_m, \quad 0 \leq m \leq \infty.$$

By the first Rogers–Ramanujan identity [71]

$$\sum_{m \geq 0} \frac{q^{m^2}}{(q; q)_m} = \frac{1}{(q, q^4; q^5)_\infty},$$

we have $\widehat{N}_C(1) = 1/(q^{-1}, q^{-4}; q^{-5})_\infty$, so $\widehat{N}_C(1)$ has an infinite product expression with modulus 5.

The state of the art of Conjecture 1 is the following theorem combining my work with Jiang [47], with Chern [20], with Jiang and Oblomkov (HJO, [48]), and with Lau and Ono [51].

Theorem 1. *Let $1 < a < b$ and $C : X^a = Y^b$ over $\mathbb{F}_q$ whose characteristic does not divide* $\gcd(a, b)$.

(a) [47, 20] *For $a = 2$, $\widehat{N}_C$ is entire. Its special values at $t = \pm 1$ have periodic product expressions, with the nontrivial evaluations governed by the Andrews–Gordon identities for odd b and the Bressoud identities for even b, of modulus $b + 2$* [2, 34, 13].

(b) [48] *For coprime a, b, the series $\widehat{N}_C(t)$ converges at $t = 1$, and*

$$\widehat{N}_C(1) = N_{a,b}(q^{-1}),$$

where $N_{a,b}(q)$ is the explicit power series in Definition 2.

(c) [51] *The series in* (b) *equals an infinite product of modulus $a + b$ proposed in* [48]*:*

$$N_{a,b}(q) = P_{a,b}(q) := \prod_{j\geq 1}(1 - q^j)^{-\rho_{a,b}(j)}, \qquad \rho_{a,b}(j) = \min\{a, b, \operatorname{dist}(aj, (a + b)\mathbb{Z})\} - 1 + \mathbf{1}_{(a+b)\mathbb{Z}}(j),$$

giving a two-parameter family of Rogers–Ramanujan-type identities indexed by coprime pairs.

Conjecture 1 remains far beyond the available methods. Even for the explicitly computable family in Theorem 1(b), the periodic product requires the deeper structures behind part (c). A central goal of my program is to identify structures that explain this rigidity when explicit computation is out of reach.

1.1. **Quot schemes of points.** Our main tool for computing $\widehat{Z}_X(t)$ is the Quot scheme. Over an arbitrary field k, it parametrizes zero-dimensional sheaves $\mathcal{Q}$ on X together with a *framing*: a surjection from a fixed coherent sheaf $\mathcal{E}$. More precisely,

$$\operatorname{Quot}_n^X(\mathcal{E}) = \{\mathcal{E} \twoheadrightarrow \mathcal{Q} : \dim_k H^0(X, \mathcal{Q}) = n,\ \dim \operatorname{Supp} \mathcal{Q} \leq 0\}/\simeq.$$

The rank-N Quot scheme refers to the case $\mathcal{E} = \mathcal{O}_X^N$, hence the rank-one Quot scheme is the classical Hilbert scheme of points. The rank-one theory already links planar singular curves to compactified Jacobians, affine Springer fibers, geometric representation theory, Catalan combinatorics, and knot homology; the last connection is marked by the celebrated Oblomkov–Rasmussen–Shende conjecture [69, 36, 38].

Definition 1. The *motivic Quot and Coh zeta functions* are

$$\mathcal{Z}_{\mathcal{E}}^X(t) = \sum_{n\geq 0}[\operatorname{Quot}_n^X(\mathcal{E})]t^n, \qquad \widehat{\mathcal{Z}}_X(t) = \sum_{n\geq 0}[\operatorname{Coh}_n(X)]t^n.$$

Their coefficients lie in the Grothendieck rings $K_0(\operatorname{Var}_k)$ of varieties and $K_0(\operatorname{Stck}_k)$ of stacks, respectively [8]. Write $\mathbb{L} = [\mathbb{A}^1]$ for the Lefschetz motive. For a reduced curve germ R with normalization $\widetilde{R}$ and a torsion-free module E of rank N, set

$$\mathcal{Z}_E^R(t) = \sum_{n\geq 0}[\operatorname{Quot}_n^R(E)]t^n, \qquad \mathcal{N}_E^R(t) = \frac{\mathcal{Z}_E^R(t)}{\mathcal{Z}_{\widetilde{R}^N}^{\widetilde{R}}(t)}.$$

Over $\mathbb{F}_q$, point counting sends $\mathbb{L}$ to q and gives the corresponding numerical series Z_E^R, N_E^R, and $\widehat{Z}_X$; the last is the series in (1).

With Jiang [47], I established three general results that underlie the sheaf counts in Theorem 1.

Theorem 2 ([47])**.**

(a) *The Coh zeta function is a rank-to-infinity limit of Quot zeta functions: for any variety X over an algebraically closed field k,*

$$\widehat{\mathcal{Z}}_X(t) = \lim_{N\to\infty}\widehat{\mathcal{Z}}_{X,N}(t), \qquad \widehat{\mathcal{Z}}_{X,N}(t) := \mathcal{Z}_{\mathcal{O}_X^N}^X(\mathbb{L}^{-N}t),$$

with coefficientwise convergence in the dimension completion [8].

(b) (Rationality) *For a reduced curve germ R over an algebraically closed field, and a torsion-free module E of rank N, the normalized Quot zeta function is a polynomial in t:*

$$\mathcal{N}_E^R(t) \in K_0(\mathrm{Var}_k)[t].$$

Moreover, $\mathcal{N}_E^R(1)$ depends only on R and N.

(c) (Functional equation) *For a reduced planar curve germ R over $\mathbb{F}_q$, the point-counting normalized Quot series $N_{R^N}^R(t)$ satisfies a functional equation generalizing the Hasse–Weil functional equation to arbitrary rank.*

Theorem 1(a)–(b) is then obtained as the rank-to-infinity limit of explicit Quot calculations.

Rationality and the functional equation come from different sources: a new geometric parametrization of submodules, and harmonic analysis in the spirit of Tate's thesis, respectively. The latter connects our theory to lattice zeta functions [73, 14, 80]. Combining explicit computations with the functional equation can establish q-series identities before a direct combinatorial proof is known. For $X^2 = Y^{2m}$, this gives a reflection identity for deformed Bressoud sums, subsequently proved by q-series methods by Chern [19, Theorem 1.8].

The quadratic family thus recovers classical Andrews–Gordon and Bressoud identities. For general coprime a, b, our work HJO [48] produces new sums directly from the geometry. Here is their explicit definition.

Definition 2. For coprime $1 < a < b$, let $G = \mathbb{N} \setminus \langle a, b\rangle$ be the set of semigroup gaps, $f = \max G = ab-a-b$, and $\delta = |G| = (a-1)(b-1)/2$. For $N \in \mathbb{N} \cup \{\infty\}$, define the generalized q-multinomial coefficient

$$\begin{bmatrix} N \\ \mathbf{n} \end{bmatrix}_{q;G} = \frac{(q;q)_N}{(q;q)_{N-n_f}} \prod_{g\in G} \frac{(q;q)_{n_g - n_{g-a-b}}}{(q;q)_{n_g-n_{g-a}}(q;q)_{n_g-n_{g-b}}}.$$

Its value is zero unless $0 \le n_g \le n_h \le N$ whenever $g, h \in G$ and $h - g \in \langle a, b\rangle$. Here $n_j = 0$ for $j \notin G$. Set

$$Q(\mathbf{n}) = \sum_{g,h\in G} K(h-g) n_g n_h, \qquad K(d) = \mathbf{1}_{d\ge 0} - \mathbf{1}_{d\ge a} - \mathbf{1}_{d\ge b} + \mathbf{1}_{d\ge a+b},$$

and, writing $|\mathbf{n}| = \sum_{g\in G} n_g$, define for $N \in \mathbb{N} \cup \{\infty\}$:

$$N_{a,b;N}(q,t) = \sum_{\mathbf{n}\in\mathbb{N}^G} \begin{bmatrix} N \\ \mathbf{n} \end{bmatrix}_{q;G} q^{Q(\mathbf{n})} t^{|\mathbf{n}|}, \qquad N_{a,b}(q) = N_{a,b;\infty}(q,1).$$

For finite N, the generalized q-multinomial lies in $\mathbb{N}[q]$ [48]. For $\mathbf{n}$ that contributes nontrivially to the sum, I proved $Q(\mathbf{n}) \ge n_f^2/\delta$ [46]. Consequently, $N_{a,b;N} \in \mathbb{N}[q,t]$ for finite N, while $N_{a,b;\infty} \in \mathbb{N}[[q,t]]$ and its specialization $N_{a,b}(q) \in \mathbb{N}[[q]]$ are well-defined formal series.

Theorem 3 ([48])**.** *Let $1 < a < b$ be coprime, $R = k[[T^a, T^b]]$, and $\widetilde{R} = k[[T]]$. With the notation above,*

(a) *For all $n, N \ge 0$, $\mathrm{Quot}_n^R(\widetilde{R}^N)$ decomposes into finitely many locally closed affine spaces, and the normalized motivic series is*

$$\mathcal{N}_{\widetilde{R}^N}^R(t) = \mathbb{L}^{N^2\delta} t^{N\delta} N_{a,b;N}(\mathbb{L}^{-1}, t^{-1}).$$

(b) *Over $k = \mathbb{F}_q$, define the finite-rank approximation to $\widehat{N}_R(t)$ by*

$$\widehat{N}_{R,N}(t) = N_{R^N}^R(q^{-N}t) = \frac{Z_{R^N}^R(q^{-N}t)}{Z_{\widetilde{R}^N}^{\widetilde{R}}(q^{-N}t)}.$$

Then $\widehat{N}_{R,N}(1) = N_{a,b;N}(q^{-1}, 1)$. Moreover, $\widehat{N}_R(t)$ converges at $t = 1$, and

$$\widehat{N}_R(1) = \lim_{N\to\infty} \widehat{N}_{R,N}(1) = N_{a,b}(q^{-1}).$$

The scaling action $T \mapsto \lambda T$ gives a Białynicki–Birula decomposition of a related moduli space into "attracting loci". Because the moduli space is singular, one cannot apply the classical theory [10] to describe these loci. Instead, a coordinate-free Gröbner construction expresses these loci through lifting problems for

representations of a two-dimensional grid poset. An acyclicity theorem removes the obstructions, giving affine fibrations over poset flag varieties. Although the full Quot series for R^N remains inaccessible in general, the functional equation and module independence in Theorem 2(b)–(c) transfer the computable $\widetilde{R}^N$ geometry to the special value in Theorem 3(b).

1.2. **Rogers–Ramanujan identities from the geometry.** The product evaluation in Theorem 1(c) follows from a stronger identity at every finite rank. It identifies the geometric sum with a different combinatorial model.

Theorem 4 (Finite identity; with Lau and Ono [51])**.** *For every coprime pair $1 < a < b$ and every $N \geq 0$,*

$$N_{a,b;N}(q,1) = (q;q)_N \, C_{\mathbf{c}_{a,b},\leq N}(q),$$

where $C_{\mathbf{c}_{a,b},\leq N}$ is the weight generating function for cylindric partitions of an explicit "balanced profile" $\mathbf{c}_{a,b}$ with entries bounded by N.

Cylindric partitions are arrays of integers weakly decreasing along rows and columns, subject to a periodic boundary condition. Passing to unbounded entries and using their known product formula [12, 30] gives $N_{a,b} = P_{a,b}$. Thus the proof discovers a finite structure underlying the infinite identities. Its main step is a common q-difference equation, reached through compositional Dyck paths, symmetric functions, slope operators, and the compositional rational shuffle theorem [9, 65]. The identity also proves a positivity conjecture of Corteel–Warnaar for our balanced coprime profiles [77].

Question 1 (Beyond torus-knot singularities)**.** What geometric structures account for the cylindric partitions and compositional Dyck paths in these formulas, and what combinatorial models govern higher-rank Quot schemes on more general singularities?

A natural next step is to extend the methods of HJO [48] to torus links $X^a = Y^b$ and the unibranched curves $k[[T^a, T^b + T^{b+1}]]$ with $\gcd(a,b) > 1$. Rank-one work already connects these settings to non-coprime rational Catalan combinatorics [37, 39]; higher rank can detect further enrichments.

1.2.1. *My experience of human-AI collaboration.* This theorem grew out of a seven-month project with three successive papers. With Lau, Ono, and Paule, I first proved four cases beyond the classical $a = 2$ family, using A_2 identities [52, 3]; Lau and Ono then proved the whole $a = 3$ family [59]; finally, our work [51] reached every coprime pair. The question and its geometric framework emerged from my work with Jiang and Oblomkov. We were developing a new problem without mature literature and known proofs to imitate. The successive proofs required new intermediate statements and changes of approach.

Throughout this process, our mathematical judgment set the direction: choosing the structures to investigate, formulating intermediate claims, and interpreting their significance. AI assistance expanded the calculations and reformulations we could explore. In the later stages, AxiomProver also formalized the arguments in Lean; the October 7 revision of [51] also formalizes the two earlier literature prerequisites, giving an unconditional Lean formalization of the finite identity and the HJO conjecture. This experience showed me how sustained interaction between mathematicians and AI can expand the scope of a research program, with the choice of questions and the development of mathematical explanations remaining central.

1.3. **Topology and arithmetic of matrix-point spaces.** Matrix points map to $\mathrm{Sym}^n(X)$ by their eigenvalues, but singular fibers remember Jordan blocks. With Asvin G, Jiang, and Wei [31], I show that their ordinary rational cohomology nevertheless admits a uniform description analogous to Macdonald's formula [63]. The explanation comes from an auxiliary space $S_n(X)$ of diagonalized matrix points, defined below.

Theorem 5 ([31])**.** *For every complex variety X, the natural map $S_n(X) \to C_n(X)$ induces an isomorphism of rational cohomology rings. In particular, writing $h^i(X) = \dim H^i(X;\mathbb{Q})$ and $P_u(X) = \sum_i (-u)^i h^i(X)$, we have*

$$\sum_{n\geq 0} \frac{P_u(C_n(X))}{(u^2;u^2)_n} t^n = \prod_{i\geq 0,\, j\geq 0} \left(\frac{1}{1-u^{i+2j}t}\right)^{(-1)^i h^i(X)}.$$

For $X = \mathbb{G}_m = \mathbb{A}^1 \setminus \{0\}$, we have $C_n(X) = \mathrm{GL}_n$ and $P_u(X) = 1 - u$. The theorem specializes to

$$\sum_{n\geq 0} \frac{P_u(\mathrm{GL}_n(\mathbb{C}))}{(u^2;u^2)_n} t^n = \frac{(ut;u^2)_\infty}{(t;u^2)_\infty} = \sum_{n\geq 0} \frac{(u;u^2)_n}{(u^2;u^2)_n} t^n,$$

where the last equality is Cauchy's q-binomial theorem. Thus

$$P_u(\mathrm{GL}_n(\mathbb{C})) = \prod_{j=1}^{n} (1 - u^{2j-1}),$$

recovering the classical formula for $P_u(U(n)) = P_u(\mathrm{GL}_n(\mathbb{C}))$ from the single input $P_u(\mathbb{G}_m)$.

The auxiliary space is

$$S_n(X) = \{((x_1, L_1), \ldots, (x_n, L_n)) : x_i \in X,\ \dim L_i = 1,\ \mathbb{C}^n = L_1 \oplus \cdots \oplus L_n\}/S_n.$$

Forgetting the diagonalization gives the map to $C_n(X)$. It is generally neither surjective nor injective: some matrix points are not diagonalizable, while repeated eigenvalues allow multiple diagonalizations. Yet it induces an isomorphism on rational cohomology. The proof compares derived direct images over $\mathrm{Sym}^n(X)$. Two key inputs are base change for good moduli space morphisms [57] and contractibility of the local nilpotent matrix-point spaces. These turn a comparison of fibers into a global cohomological statement.

The arithmetic behavior is subtler, but point counts also exhibit statistical regularity.

Theorem 6 (with Ono and Saad [54])**.** *Fix $n, r \geq 1$. As X varies over the standard affine models of the Legendre elliptic family or the $K3$ family studied by Ahlgren–Ono–Penniston* [1]*, the matrix-point counts $|C_n(X)(\mathbb{F}_{p^r})|$, after explicit centering and rescaling, converge to explicit Sato–Tate-type distributions as $p \to \infty$. In the elliptic case, the limiting law is the semicircle distribution.*

The subsequent REU project that I co-mentored establishes parallel results for fixed elliptic curves and $K3$ surfaces over $\mathbb{Q}$, with the prime varying [11].

Question 2 (A cohomological explanation of point counts)**.** Can a cohomological theory of matrix points explain both the uniformity of ordinary cohomology and the finer arithmetic structure of point counts?

The usual trace formula uses compactly supported cohomology, so Theorem 5 does not immediately answer this question. Understanding what ordinary cohomology forgets about the singular fibers could also help explain the rigidity predicted by Conjecture 1.

1.4. **Colored knot homology conjectures.** Our computations in HJO suggest a higher-rank extension of the Oblomkov–Rasmussen–Shende picture [48, 69]. The framing rank becomes a knot-theoretic color: the one-row representation S^N. A second geometric model encodes the same integer as the multiplicity of a nonreduced curve.

Conjecture 2 (Quot schemes, thickened curves, and colored knots [48])**.** *Let $R = \mathbb{C}[[X,Y]]/(f)$ be a reduced unibranched planar curve germ, let K be its algebraic knot, and let $N \geq 1$. After normalization and explicit changes of grading variables, there is a three-way identification among*

(a) *the motivic generating series of* $\mathrm{Quot}_n^R(\widetilde{R}^N)$*;*
(b) *the motivic generating series of* $\mathrm{Hilb}_n(\mathbb{C}[[X,Y]]/(f^N))$*;*
(c) *a two-variable specialization of the bottom row of reduced S^N-colored HOMFLY homology of K.*

Here the bottom row is the part of lowest HOMFLY degree, in the grading conventions of Gorsky–Gukov–Stošić [35]*. We also conjecture that the trigraded bottom row is recovered by the perverse Leray filtrations on the two geometric models, and a different specialization gives the normalized Quot series for R^N.*

At $N = 1$, the geometric comparison (a)(b) becomes the Hilb-vs-Quot conjecture of Kivinen–Trinh [58].

Several branches. In our new preprint *Sheaves on nodal singularities I* [49], Jiang, Oblomkov, and I compute motivic generating functions for punctual Hilbert schemes of $X^u Y^v = 0$ and punctual Quot schemes on the node $XY = 0$ with independent framing ranks on its two branches. The formulas prove the Hilb–Quot correspondence for the node. We also formulate a branchwise Hilb–Quot–HOMFLY conjecture and test it on the colored Hopf link.

2. Topology of configuration spaces

The *unordered configuration space* parametrizes unordered tuples of distinct points:

$$\mathrm{Conf}_n(X) = \{(x_1, \dots, x_n) \in X^n : x_i \neq x_j \text{ for } i \neq j\}/S_n.$$

For $X = \mathbb{C}$, its topology is that of the braid group, going back to Arnol'd [4].

Theorem 7 (Configuration spaces of punctured surfaces; with Ramos [55])**.** *Let $X = \Sigma_{g,r}$, an r-punctured Riemann surface of genus g. If $r > 0$, the mixed Hodge numbers of its unordered configuration spaces are given by an explicit rational generating function:*

$$f_X(x,y,u,t) \coloneqq \sum_{p,q,i,n\geq 0} (-1)^i h^{p,q;i}(\mathrm{Conf}_n(X)) x^p y^q u^i t^n = \frac{\Phi_g((1-xyu^2t^2)(1-xut)^g(1-yut)^g)}{(1+xyut)^{r-1}(1-t)(1-x^2yu^2t^2)^g(1-xy^2u^2t^2)^g},$$

where Φ_g shifts the u-degree down by 1 for monomials with u-degree at least $g+2$, and fixes other monomials.

My earlier work with Cheong proves the case $g = r = 1$ [16]; my separate work describes the effect of removing further points under a mixed-Hodge theoretic hypothesis satisfied by punctured Riemann surfaces [43]. Theorem 7 gives the mixed-Hodge refinement of the Betti-number formulas of Drummond-Cole–Knudsen [26], and packages them in a single rational expression with a clean numerator.

The proof uses Totaro's spectral sequence [75]. The special feature here is that the differential can be algebraically identified as a Hard Lefschetz map for an abelian variety of dimension g. The transition from injectivity to surjectivity explains the piecewise shift Φ_g.

Representation stability regulates the behavior of $H^i(\mathrm{Conf}_n(X))$ as n varies with i fixed, while higher-order variants examine certain linear slices in (n, i) [23, 24, 66]. The rationality of f_X suggests asking for a structure that organizes all cohomological degrees and numbers of points together.

Question 3 (Rationality and its explanation)**.** For which varieties X is f_X rational? What geometric or representation-theoretic structures explain its numerator and organize cohomology as both the number of points and cohomological degree vary?

Even for spaces where this question is answered, a finer question can still give further insights.

Question 4 (Equivariant version)**.** What natural class of symmetric-function series contains the Frobenius-character generating functions for cohomology of *ordered* configuration spaces, with the natural S_n-action? When do these series admit product formulas or analogues of rationality?

For $X = \mathbb{R}^d$, $d \geq 2$ (including $\mathbb{C} = \mathbb{R}^2$), the S_n-representations are known [60, 74], and their Frobenius-character series admit product formulas [41, Section 2.7]. Punctured surfaces of positive genus offer a next setting in which to seek comparable formulas.

3. Discrete random matrices and Cohen–Lenstra distributions

The Cohen–Lenstra heuristics weight a random algebraic object, such as a finite abelian group, by the inverse size of its automorphism group [25]. Random integral matrices produce random abelian groups by taking cokernels, and different models lead to different Cohen–Lenstra-type distributions: general matrices [79], symmetric matrices [78], polynomials of matrices [18], and matrix products [68]. I seek enriched structures that make the laws simple. My contributions concern the following two settings.

For independent random matrices $M_1, \dots, M_k$, the joint distribution of the groups $\mathrm{cok}(M_1 \cdots M_i)$, $1 \leq i \leq k$, was determined by Nguyen and Van Peski [68]. Its Cohen–Lenstra structure becomes transparent when one also retains the natural quotient maps.

Theorem 8 (Flag universality for matrix products [45, 53])**.** *For independent Haar-random p-adic $n \times n$ matrices, the random* flag *of abelian groups*

$$\mathrm{cok}(M_1 \cdots M_k) \twoheadrightarrow \cdots \twoheadrightarrow \mathrm{cok}(M_1)$$

with natural quotient maps obeys the finite-rank Cohen–Lenstra distribution [45]*. For fixed k, its $n \to \infty$ limit assigns a finite flag $\mathbf{G}$ probability*

$$\frac{1}{|\mathrm{Aut}(\mathbf{G})|} \prod_{j \geq 1} (1 - p^{-j})^k,$$

where automorphisms are required to commute with quotient maps. With Nguyen and Van Peski [53]*, I prove that the same limit holds when all entries are independent and balanced modulo p: each residue class has probability at most $1 - \varepsilon$, for some fixed $\varepsilon > 0$.*

A different regime arises when the matrix has a prescribed residue modulo p, so the balancedness hypothesis fails.

Theorem 9 (with Cheong [17])**.** *For a nonconstant monic $P(t) \in \mathbb{Z}_p[t]$ and a Haar-random p-adic matrix M with prescribed residue modulo p, the exact distribution of $\mathrm{cok}(P(M))$ as a finite module over*

$$R = \mathbb{Z}_p[t]/(P(t))$$

is an explicit Cohen–Lenstra-type law: on its support, a module G has weight proportional to $1/|\mathrm{Aut}_R(G)|$, with G/pG prescribed as an R/pR-module.

We also give counterexamples to unrestricted universality in this fixed-residue setting [17, Remark 1.12].

Question 5. For a fixed finite collection of associative polynomials in random matrices, what is the limiting joint cokernel distribution? When does additional algebraic structure make that law Cohen–Lenstra-type, and when is the law universal?

Work in progress. With Nguyen, I am studying a key family of cases where limiting moments can be obtained by methods in [53].

Work in progress. With Lvov, Van Werde, and Zhang, I am studying cokernels of polynomials in random *symmetric matrices*, including the additional structure needed to describe their distribution.

4. Rigidity in arithmetic geometry

4.1. **Low-degree subvarieties of universal hypersurfaces.** How can one choose d unordered points on each degree-d plane curve, with the choice varying algebraically over a dense open subset of the parameter space? A simple construction is intersecting with a line that varies with the curve. Ma shows that every such family arises this way [61, Theorem 17]. With Kadets and Martin, I extend this rigidity to cycles of arbitrary dimension on sufficiently high-degree hypersurfaces [50].

Theorem 10 ([50])**.** *For $n, d \geq 1$, let K be the function field of the space of degree-d hypersurfaces of $\mathbb{P}^{n+1}_{\mathbb{C}}$ and $X \subseteq \mathbb{P}^{n+1}_K$ be the universal hypersurface.*

(a) *The degree of every cycle on X is divisible by d.*
(b) *If $k \geq 1$ and $d \geq 4kn + 3k^3 - k^2 + 1$, every effective cycle Z of dimension $m \geq 0$ and degree kd on X is the intersection cycle $X \cdot W$ for an effective K-cycle W of dimension $m + 1$ and degree k in $\mathbb{P}^{n+1}_K$, with no component contained in X.*

This answers questions of Farb and Ma [29, 61, 62]. The proof combines the Riedl–Yang Grassmannian method [70], Mumford–Roitman results on zero-cycles [67, 72], and our analysis of Cayley–Bacharach conditions under a Galois action [50, Section 4], inspired by their use in measures of irrationality [5, 6]. A lifting theorem of Chiantini–Ciliberto [21] passes from zero-cycles to higher-dimensional cycles.

Question 6. Can we replace $\mathbb{P}^n$ by other ambient varieties, and sufficiently high-degree hypersurfaces by "sufficiently ample" linear systems?

I expect the degree threshold to be replaced by a convex region in the ample cone that is stable under dilation by every factor $c \geq 1$. Products of projective spaces and, more generally, toric varieties give concrete settings in which to formulate and test such an extension.

4.2. **Noncommutative Mordell–Lang problems.** The classical Mordell–Lang theorem states that, in characteristic zero, the intersection of a subvariety of a semiabelian variety with a finitely generated subgroup is a finite union of cosets of subgroups [28, 76]. Semi-abelian varieties are commutative; I study analogues in noncommutative algebras.

Theorem 11 (with Bell and Ghioca [33, 7])**.** *Let A be a finite-dimensional central simple algebra over a field K of characteristic zero, let $V \subseteq A$ be a closed K-subvariety, and let $f_1, \dots, f_r \in A^\times$. Set*

$$S = \{(n_1, \dots, n_r) \in \mathbb{Z}^r : f_1^{n_1} \cdots f_r^{n_r} \in V\}.$$

(a) *If A is a division algebra, then S is a finite union of cosets of subgroups of $\mathbb{Z}^r$* [33].
(b) *If A is a division algebra, $0 \notin V$, and $f_1, \dots, f_r$ have multiplicatively independent norms, then S is finite* [33].
(c) *If $A = \mathrm{Mat}_\ell(K)$, $0 \notin V$, and the sets of distinct eigenvalues of the f_i in $\overline{K}$ are pairwise disjoint with multiplicatively independent union, then S is finite* [7].

My starting point was the following conjecture, which I formulated in my work [42] on quaternionic unit equations.

Conjecture 3 (Noncommutative unit equations [42, Conjecture 1.4])**.** *Let D be a finite-dimensional division algebra over $\mathbb{Q}$, let $\Gamma_1, \Gamma_2 \subseteq D^\times$ be finitely generated subsemigroups and $a, b, c, d \in D^\times$ be fixed. Then*

$$axb + cyd = 1, \qquad x \in \Gamma_1, \quad y \in \Gamma_2,$$

has only finitely many solutions.

If D is commutative, this is the classical S-unit theorem [27]. I proved the first noncommutative case:

Theorem 12 ([42])**.** *Let D be the algebra of Hamilton quaternions with coordinates in $\overline{\mathbb{Q}} \cap \mathbb{R}$, and let $f_1, \dots, f_r, g_1, \dots, g_s \in D^\times$ have Euclidean norms greater than 1. Let Γ_1 be the subsemigroup generated by $f_1, \dots, f_r$, and set*

$$\Gamma_2 = \{g_1^{n_1} \cdots g_s^{n_s} : n_1, \dots, n_s \in \mathbb{Z}_{\geq 0}\}.$$

If the g_j commute pairwise, then Conjecture 3 is true in this setting.

This has applications to arithmetic dynamics [42, Corollary 1.6].

A variant of Theorem 11(b) for nonnegative exponents yields the following extension.

Theorem 13 (with Ghioca [33])**.** *In the setting of Theorem 12, the finiteness still holds without the commutativity assumption on g_j.*

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